

Extended Abstract

Introduction

The first wave of Pokemon introduced in 1996 would revolutionize the card-game industry for years to come and attract over 30 million new players to the franchise. As someone who grew up inside the Pokemon bubble I wanted to investigate some of the design choices Game Freak used when it came to building out the first generation of Pokemon. I used the MondoDB document store model to analyse the type distribution as well as other important factors in the dataset.

Analysis

To begin my analysis I retrieved my data from the public PokeAPI (pokeapi.co) using Python's request library to extract the JSON documents needed. After which I had imported each document into MongoDB as its own collection. Each document was set as an array field allowing me to use \$unwind and \$group aggregations to count occurrences across the entire collection revealing patterns in the data.

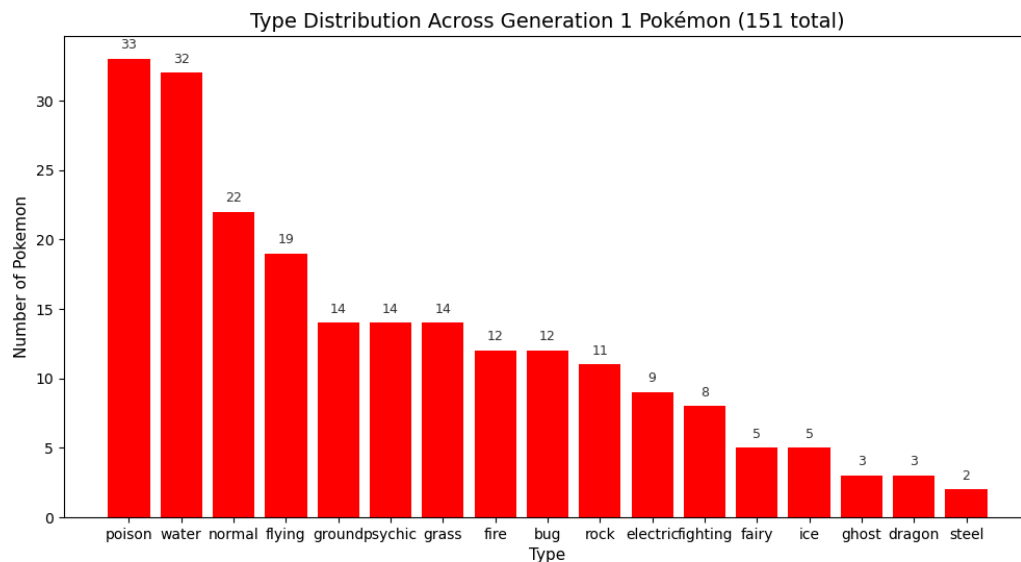
After running many of my queries it was surprising to find out there were 33 Poison types closely followed by water at 32. This challenged the common misconception that water is the most dominant type in the first generation that Pokemon released. Although a closer look at the data would reveal that many Pokemon such as Weedle, Ekans, and various Grass-types carry Poison as a secondary type which might inflate the numbers a little.

Conclusions

Overall this report exposed me to the fact that MongoDB's aggregation pipeline is well suited for explanatory analysis and the lack of constraints proved more helpful than a perceived drawback.

Specifically the \$unwind function allowed for type level analysis which would require a JOIN in a relational database. In conclusion MongoDB is very useful and I would be incorporating it into many of my future projects.

Visualization



References

PokeAPI. (2024). *The RESTful Pokémon API*. <https://pokeapi.co>

MongoDB, Inc. (2024). *MongoDB Documentation: Aggregation Pipeline*.
<https://www.mongodb.com/docs/manual/core/aggregation-pipeline/>